

Parts Inquiry & SAP Business One PO API Integration

Extended Integration Guide: Normal Stock Orders (DS / SO), Emergency Orders (EO), and Automated Hybrid Routing

TARGET AUDIENCE	PREPARED BY	CORE SYSTEMS	DOCUMENT VERSION
ABS (SAP B1 Technical Team)	Motasem Ghanem (EQP Ops)	EQP Portal • Komatsu PDX • SAP B1	v2.0 (Hybrid DS/SO + EO RFC)

1. Executive Summary & Business Objective

Dar Al Hai is the authorized Komatsu equipment operator and service organization managing an extensive fleet of excavators, bulldozers, and heavy earthmoving machinery across major infrastructure projects in Kuwait. To maintain operational readiness and rapidly replenish replacement components, the Maintenance & Spare Parts department orders directly from **Komatsu Middle East (KME / PDX Portal)**.

The system now supports **Dual-Channel Procurement**:

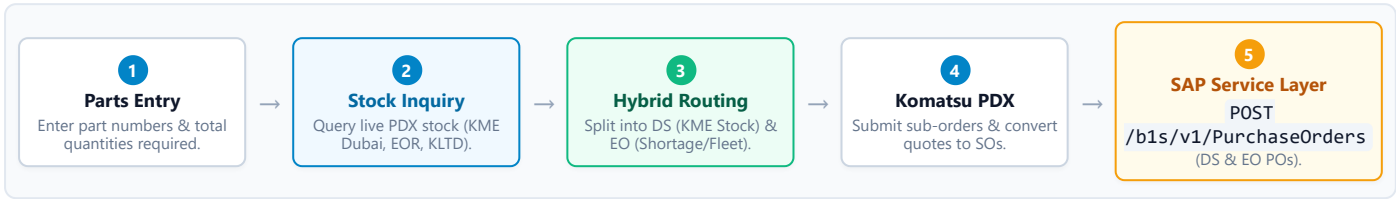
- Normal Stock Orders (DS / SO)**: Used for routine stock replenishment when components are physically in stock at KME Dubai warehouse. These orders carry **0.00% surcharge**, use **DDU** terms, and require **no machine serial numbers or customer assignments**.
- Emergency Orders (EO)**: Used for urgent breakdown recovery where parts are restricted or backordered. These orders require **13.30% surcharge**, use **EXW** terms, and strictly mandate **genuine machine model & serial binding** with line-item caps.
- Automated Hybrid Splitting**: When an order contains high quantities, the EQP system automatically splits requested parts between available KME stock (DS) and breakdown shortages (EO).

The Objective for the ABS (SAP Team):

We need to enable the EQP system to automatically create corresponding **Purchase Orders (POs)** in **SAP Business One** for Vendor **V000006 (Komatsu Middle East)** via the **SAP B1 Service Layer (REST API)**. Both Stock Orders (DS / SO) and Emergency Orders (EO) must be recorded in SAP B1 with the appropriate commercial terms, surcharges, and machine metadata.

2. End-to-End System Architecture & Hybrid Flow

The updated operational flow connects field requisitions directly to SAP Business One through an intelligent stock-routing engine:



3. Order Types & Commercial Policies Matrix (Essential for SAP)

Komatsu PDX enforces radically different commercial terms and data requirements depending on the order type. SAP Business One must record each PO according to its specific classification:

FEATURE / DIMENSION	NORMAL STOCK ORDERS (DS / SO)	EMERGENCY ORDERS (EO)
Definition & Purpose	DS: Daily Stock Shipment • SO: Stock Order. General warehouse replenishment.	EO: Emergency Order. Urgent breakdown recovery for halted equipment.
Fulfilled From	KME Stock (Dubai local warehouse physical stock).	KME EOR (Dubai restricted) or KLTD (Japan factory).
Premium Surcharge	0.00% Premium (Base DNet catalogue pricing).	+13.30% Emergency Surcharge added to total line value.
Incoterms / Delivery	DDU (Delivery Duty Unpaid — Delivered to Kuwait).	EXW (Ex Works — Picked up at KME PDC Dubai).
Machine Model & Serial	NOT REQUIRED (Model: Blank, Serial: Blank). Komatsu PDX disables machine fields.	STRICTLY MANDATORY (e.g. Model PC500LC-10R , SN 100433). Chassis validated against Komatsu worldwide database.
Customer Account	NOT REQUIRED / BLANK (General Stock). Order belongs to dealership distributor account.	MANDATORY (e.g. LAALA AL KUWAIT REAL ESTATE CO.).
Line Item Caps	NO CAP PER ITEM (e.g. 50 or 500 units allowed on 1 PO).	STRICTLY CAPPED (e.g. max 1, 2, or 5 EA per machine asset).
SAP Receiving Warehouse	Central General Spare Parts Warehouse (e.g. 001 / 003).	Project / Job-Site Breakdown Warehouse (e.g. 003 / Project).

4. Real-World Hybrid Ordering Scenario: 50 Units Requested

Consider an engineer requesting **50 units of Fuel Filter (Part 600-311-6742)**:

The Problem if Ordered Purely via Emergency Channel (EO):

Komatsu caps Fuel Filters at **20 units per machine** on EO. Furthermore, placing 50 units on EO incurs an unnecessary **13.30% premium surcharge** across the entire batch, even on parts already sitting in Dubai!

The Intelligent EQP Hybrid Solution:

The EQP system checks live PDX availability:

- **KME Stock available:** 10 units.
- **Remaining shortage:** 40 units (factory backorder).

Automated Order Generation:

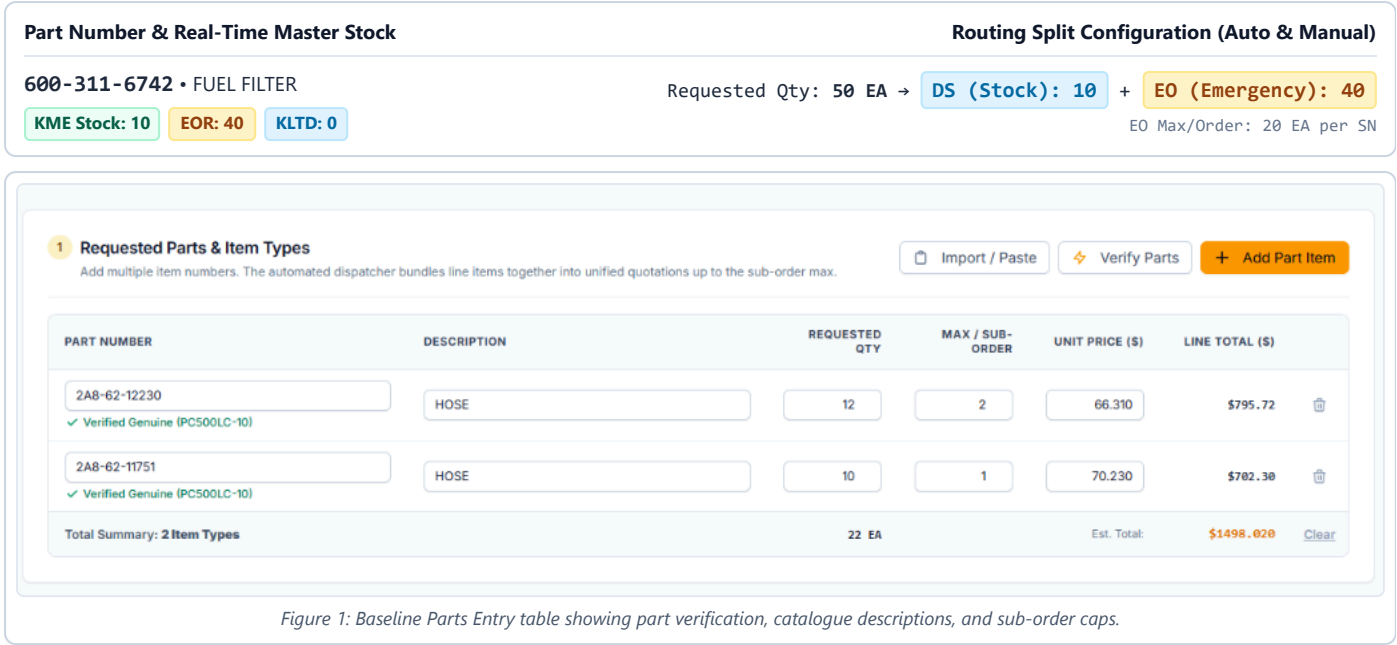
1. **Sub-Order #1 (**DS**)** — Ref **R229/2026** : 10 units routed to Daily Stock. **0% premium, DDU, no machine serial, no customer name**. Consolidates into **1 PO in SAP**.
2. **Sub-Order #2 (**EO**)** — Ref **R230/2026** : 20 units routed to Machine #1 (Model **PC500LC-10R**, SN **100433**). 13.3% premium, EXW. Generates **1 PO in SAP**.
3. **Sub-Order #3 (**EO**)** — Ref **R231/2026** : 20 units routed to Machine #2 (Model **PC500LC-10R**, SN **100434**). 13.3% premium, EXW. Generates **1 PO in SAP**.

Financial Benefit: Dar Al Hai saves the 13.3% surcharge on the 10 local units and ensures 100% portal compliance without rejections.

5. Step-by-Step Functional Workflow & UI Verification

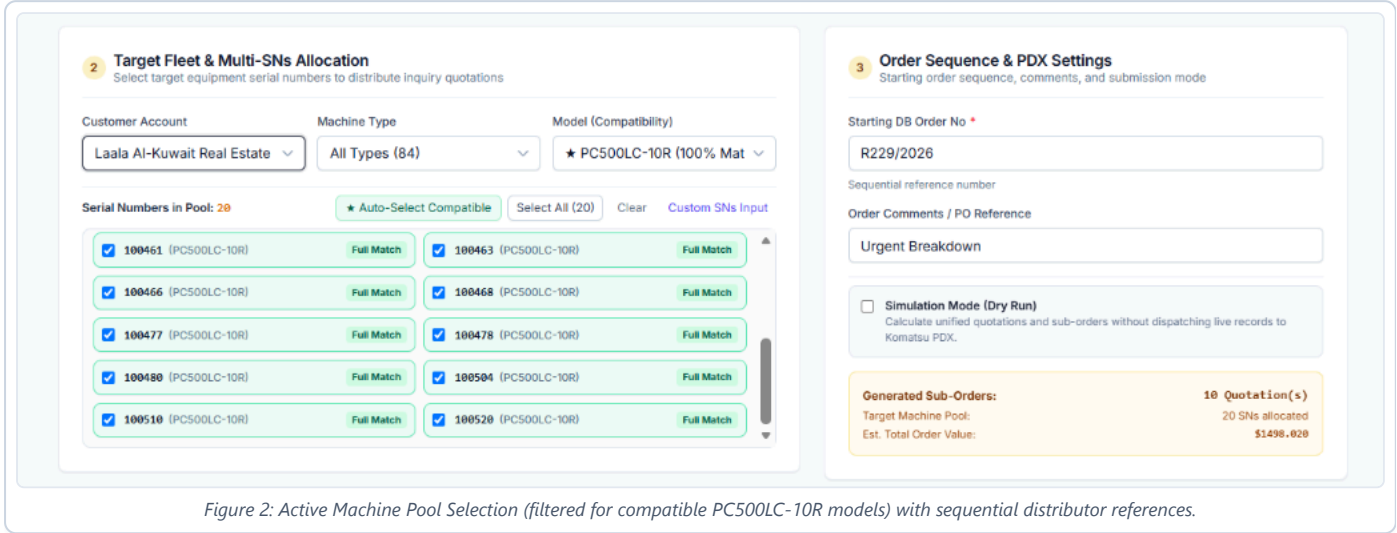
Step 1: Part Entry, Real-Time Stock Badges & Routing Split Inputs

The maintenance dispatcher enters part numbers. The system instantly queries the Komatsu PDX master and renders real-time stock availability badges: **KME Stock** (Dubai), **EOR** (Emergency Restriction), and **KLTD** (Factory). The user can toggle between **DS (Daily)** and **SO (Stock)** as the default normal order type.



Step 2: Intelligent Fleet Allocation & 100% Stock Bypass

If an order is **100% Stock** (**DS** / **SO**), Step 2 automatically displays a "Machine Details Not Required — Bypassed" banner and skips fleet selection entirely. If the order is **Hybrid**, the fleet selector is activated *strictly for the EO portion* (40 units divided across 2 machines).



Step 3: Unified Quotations Manifest (Clear Stock vs Emergency Visual Badging)

The EQP engine generates the unified manifest, clearly differentiating normal stock from emergency breakdown sub-orders:

- **Sub-Order #1:** Displays **DS • Direct Stock** badge, Asset: "KME Direct Stock • No Machine Required • DDU", uncapped.
- **Sub-Order #2 & #3:** Displays **EO • Emergency** badge, Asset: "SN: 100433 • PC500LC-10R • Laala Al-Kuwait", capped at 20 EA.
- **Continuous Distributor Sequence:** R229/2026 → R230/2026 → R231/2026 .

4 Planned Unified Quotations & Live Dispatch Queue								
10 sub-orders generated across 2 requested parts (22 units)								
			Export CSV		Export SAP Excel		Dispatch All (10)	
#	DB ORDER NO	TARGET ASSET	BUNDLED PARTS REQUESTED		TOTAL QTY	EST. VALUE (\$)	QUOTATION #	ACTION
1	R229/2026	100433 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA	2A8-62-11751 : 1 EA	3	\$202.850	—	Ready Inspect
2	R230/2026	100434 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA	2A8-62-11751 : 1 EA	3	\$202.850	—	Ready Inspect
3	R231/2026	100439 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA	2A8-62-11751 : 1 EA	3	\$202.850	—	Ready Inspect
4	R232/2026	100444 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA	2A8-62-11751 : 1 EA	3	\$202.850	—	Ready Inspect
5	R233/2026	100447 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA	2A8-62-11751 : 1 EA	3	\$202.850	—	Ready Inspect
6	R234/2026	100448 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA	2A8-62-11751 : 1 EA	3	\$202.850	—	Ready Inspect
7	R235/2026	100449 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA		1	\$70.230	—	Ready Inspect
8	R236/2026	100454 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA		1	\$70.230	—	Ready Inspect
9	R237/2026	100456 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA		1	\$70.230	—	Ready Inspect
10	R238/2026	100459 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA		1	\$70.230	—	Ready Inspect
View Technical Dispatch Audit Logs (4 entries)								

Figure 3: Planned Unified Dispatch Queue showing sub-order references, target asset serial numbers, and line items.

Step 4: Komatsu Quotation to Sales Order (SO) Conversion & SAP Trigger

Once submitted to Komatsu PDX, the portal returns official Komatsu Quotation Numbers (e.g. 0000282871). The EQP system batch-confirms these quotations and converts them into official **Komatsu Sales Orders (SO)** (e.g., SO #0000278046 , SO #0000278047 ...).

Komatsu Quotation to Sales Order (SO) Converter

Batch confirm in-process quotations and copy them directly to Komatsu Sales Orders

Refresh

1-Click Convert (0)

Create SAP PO (0)

Export SAP Excel (0)

All (25)

Copied to SO Only (25)

Selected: 0 of 25

☐	QUOTATION #	DB ORDER NO	CUSTOMER	PERSON IN CHARGE	AMOUNT (USD)	STATUS	ACTIONS
☐	0000282871	R228/2026	DAR AL HAI	motasemgha	240.72	Transferred to SO ✓ SO #0000278046	Create SAP PO Unmark
☐	0000282870	R227/2026	DAR AL HAI	motasemgha	419.26	Transferred to SO ✓ SO #0000278047	Create SAP PO Unmark
☐	0000282869	R226/2026	DAR AL HAI	motasemgha	419.26	Transferred to SO ✓ SO #0000278048	Create SAP PO Unmark
☐	0000282863	225/2026	DAR AL HAI	motasemgha	1,383.84	Transferred to SO ✓ SO #0000278044	Create SAP PO Unmark
☐	0000282560	R224/2026	DAR AL HAI	mrami	549.90	Transferred to SO ✓ SO #0000277989	Create SAP PO Unmark
☐	0000282513	R223/2026	DAR AL HAI	mrami	116.00	Transferred to SO ✓ SO #0000277690	Create SAP PO Unmark
☐	0000282365	R222/2026	DAR AL HAI	mrami	142.20	Transferred to SO ✓ SO #0000277538	Create SAP PO Unmark
☐	0000282364	R221/2026	DAR AL HAI	mrami	245.08	Transferred to SO ✓ SO #0000277539	Create SAP PO Unmark
☐	0000282363	R220/2026	DAR AL HAI	mrami	245.08	Transferred to SO ✓ SO #0000277540	Create SAP PO Unmark
☐	0000282362	R219/2026	DAR AL HAI	mrami	245.08	Transferred to SO ✓ SO #0000277541	Create SAP PO Unmark
☐	0000282361	R218/2026	DAR AL HAI	mrami	245.08	Transferred to SO ✓ SO #0000277542	Create SAP PO Unmark

Figure 4: Komatsu Quotation to Sales Order Converter showing official Komatsu SO numbers and the target "Create SAP PO" action.

The Exact Integration Trigger:

When the Komatsu Sales Order is confirmed, clicking **"Create SAP PO"** calls the SAP Service Layer API to instantly create the approved Purchase Order in SAP B1 for Vendor **V000006** , returning the SAP DocNum directly into this screen.

6. Technical API Specification for the ABS / SAP Team

We propose integrating directly with the **SAP Business One Service Layer** (OData v4 REST API).

A. Header Field Mapping (Table: `OPOR` / Entity: `PurchaseOrders`)

SAP FIELD	SAP DESCRIPTION	STOCK ORDER (<code>DS</code> / <code>SO</code>)	EMERGENCY ORDER (<code>EO</code>)
<code>CardCode</code>	Vendor Code	<code>V000006</code> (Komatsu Middle East)	<code>V000006</code> (Komatsu Middle East)
<code>DocDate</code>	Posting Date	Current Date (<code>2026-10-05</code>)	Current Date (<code>2026-10-05</code>)
<code>DocDueDate</code>	Delivery Due Date	Expected Delivery Date	Expected Delivery Date (Urgent)
<code>DocCurrency</code>	Document Currency	USD	USD
<code>NumAtCard</code>	Vendor Ref / BP Ref	<code>SO #0000278046 / R229/2026</code>	<code>SO #0000278047 / R230/2026</code>
<code>Comments</code>	Remarks	Komatsu DS Stock Replenishment (KME)	Komatsu EO: PC500LC-10R (SN: 100433)
<code>U_OrderType</code>	UDF: Order Type	DS (or SO)	EO
<code>U_KomatsuSO</code>	UDF: Komatsu SO No	Official Komatsu SO Number	Official Komatsu SO Number
<code>U_DeliveryTerms</code>	UDF: Delivery Terms	DDU	EXW
<code>U_PremiumRate</code>	UDF: Surcharge Rate	<code>0.00</code> (0% surcharge)	<code>13.30</code> (13.3% surcharge)
<code>U_MachineSerial</code>	UDF: Machine Serial	<i>null / empty string</i> (Not applicable)	<code>100433</code> (Chassis Serial)
<code>U_MachineModel</code>	UDF: Machine Model	<i>null / empty string</i> (Not applicable)	<code>PC500LC-10R</code> (Model Code)
<code>U_Customer</code>	UDF: End Customer	<i>null / empty string</i> (General Stock)	<code>LAALA AL KUWAIT REAL ESTATE CO.</code>

B. Concrete JSON Payload Comparison: Normal Stock vs Emergency Breakdown

Example 1: Normal Stock PO (DS) — No Machine Details

```
{
  "CardCode": "V000006",
  "DocDate": "2026-10-05",
  "DocDueDate": "2026-10-12",
  "DocCurrency": "USD",
  "NumAtCard": "SO #0000278046 / R229/2026",
  "Comments": "Komatsu DS Stock Replenishment",
  "U_OrderType": "DS",
  "U_KomatsuSO": "0000278046",
  "U_KomatsuQuote": "0000282871",
  "U_DeliveryTerms": "DDU",
  "U_PremiumRate": 0.0,
  "U_MachineSerial": null,
  "U_MachineModel": null,
  "U_Customer": null,
  "DocumentLines": [
    {
      "ItemCode": "600-311-6742",
      "ItemDescription": "FUEL FILTER",
      "Quantity": 10.0,
      "UnitPrice": 25.00,
      "WarehouseCode": "001",
      "TaxCode": "P0"
    }
  ]
}
```

Example 2: Emergency Breakdown PO (EO) — With Machine Details

```
{
  "CardCode": "V000006",
  "DocDate": "2026-10-05",
  "DocDueDate": "2026-10-08",
  "DocCurrency": "USD",
  "NumAtCard": "SO #0000278047 / R230/2026",
  "Comments": "Komatsu EO: PC500LC-10R (SN: 100433)",
  "U_OrderType": "EO",
  "U_KomatsuSO": "0000278047",
  "U_KomatsuQuote": "0000282872",
  "U_DeliveryTerms": "EXW",
  "U_PremiumRate": 13.3,
  "U_MachineSerial": "100433",
  "U_MachineModel": "PC500LC-10R",
  "U_Customer": "LAALA AL KUWAIT",
  "DocumentLines": [
    {
      "ItemCode": "600-311-6742",
      "ItemDescription": "FUEL FILTER",
      "Quantity": 20.0,
      "UnitPrice": 28.325,
      "WarehouseCode": "003",
      "TaxCode": "P0"
    }
  ]
}
```

C. Expected SAP Response & Cross-System Traceability

Upon successful creation, the SAP B1 Service Layer returns HTTP 201 Created with document numbers:

```
{
  "DocEntry": 14205,
  "DocNum": 20260481,
  "DocDate": "2026-10-05",
  "CardCode": "V000006",
  "DocTotal": 250.00,
  "DocCurrency": "USD"
}
```

The EQP database saves `DocNum: 20260481` and `DocEntry: 14205` directly onto the Komatsu quotation record, giving maintenance, procurement, and accounting teams complete end-to-end traceability across both systems.

7. Key Questions & Alignment Needed from the ABS Team

1. Service Layer Endpoint URL & Firewall Whitelisting:

Please provide the official Service Layer URL (e.g. `https://daralhai.b1pro.com:50000/b1s/v1/`) and confirm network access for the EQP server.

2. Dedicated Technical API User:

We request a dedicated API service account (e.g. `B1_API_EQP`) with permissions restricted to `PurchaseOrders` (Create/Read) and `Items` (Read/Create).

3. Warehouse Code Mapping for Stock vs Emergency:

Should normal stock orders (`DS` / `SO`) route to the Central Stock Warehouse (e.g. `001`) while emergency breakdown parts (`EO`) route to the Job-Site/Transit Warehouse (e.g. `003`)?

4. Pricing & 13.30% Surcharge Representation:

For Emergency Orders, should the 13.30% Komatsu surcharge be incorporated into the line-item unit price (e.g. $\$25.00 \times 1.133 = \28.325) or recorded as an additional document freight/charge expense?

5. Handling New Part Numbers (Item Master `ITM`):

With over 500,000 Komatsu parts, an urgent breakdown item may not yet exist in SAP B1. What is ABS's preferred procedure?

- *Option 1 (Recommended):* Allow our API to auto-create missing parts via `POST /b1s/v1/Items` using Komatsu catalogue master data before creating the PO.
- *Option 2:* Use a generic non-inventory item master with explicit line-item descriptions.

6. PO Approval Status (Active vs Draft):

Should the API create approved open POs (`PurchaseOrders`) directly, or create Drafts (`PurchaseOrderDrafts`) during the pilot validation period?

7. Existing UDF Alignment on `OPOR`:

Please confirm if user-defined fields already exist on `OPOR` for `U_OrderType`, `U_KomatsuSO`, `U_MachineSerial`, and `U_DeliveryTerms`, or if ABS will create them.

Next Steps & Testing Plan:

Once ABS provides the Service Layer connection credentials and confirms the UDF structure, the EQP engineering team will execute sample test transactions in the SAP sandbox environment within 24 hours.